

Adaptive Kernel Selection for Stein Variational Gradient Descent

Anonymous Authors

Abstract

Efficient posterior approximation remains a central quantitative objective in Bayesian inference. Stein Variational Gradient Descent (SVGD) transports a finite particle ensemble toward a target distribution, but its convergence rate, variance estimation, and final approximation error are strongly governed by the reproducing kernel Hilbert space (RKHS) in which the update direction is chosen. The median heuristic supplies a convenient bandwidth baseline, yet its single-scale choice can be poorly matched to the empirical geometry of the particles, especially in higher dimensions. We propose Adaptive SVGD (Ad-SVG), which compares kernels within a parameterized family at each stage of inference and updates the kernel parameters by ascending the kernelized Stein discrepancy (KSD) before applying the SVGD particle step. This framing makes the quantitative objective explicit: the selected kernel is the one, up to the accuracy of the inner update, that yields the largest KSD and therefore the largest instantaneous KL decrease available within the class. We extend fixed-kernel KSD convergence arguments to this adaptive setting, proving convergence of the maximal KSD over the kernel class and giving refined KL bounds under a Stein logarithmic Sobolev inequality. Across the reported experiments, Ad-SVG improves the comparison against the median-heuristic baseline: it preserves posterior uncertainty more accurately, reduces approximation error in higher-dimensional Gaussian tests, and matches or improves predictive and discrepancy metrics in Bayesian logistic regression.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based approach to approximate Bayesian inference and is often used as a computational alternative to traditional Markov Chain Monte Carlo (MCMC) methods. Quantitatively, SVGD moves a particle set by following functional-gradient directions for the Kullback-Leibler (KL) divergence to the target distribution, with each update restricted to the unit ball of a reproducing kernel Hilbert space (RKHS). This RKHS choice is not a benign implementation detail: the kernel determines particle interactions and has a direct, measurable effect on convergence and approximation quality [8, 28]. Empirically, standard SVGD particle systems also often underestimate posterior variance [2], so comparisons in moderate and high dimensions frequently turn on how much uncertainty each method retains. These patterns motivate the central comparison in this paper. Rather than accepting the median heuristic as a fixed baseline bandwidth rule, we adapt the kernel parameters during inference by maximizing the kernelized Stein discrepancy (KSD), so that the SVGD update is paired with the kernel that gives the strongest discrepancy signal available within the chosen class.

1.1 Related work.

Since its introduction [18], SVGD has been evaluated and deployed across a broad set of approximate-inference tasks in machine learning [22, 24, 30, 15]. Theoretical work has clarified which convergence

quantities are controlled by the method: mean-field convergence has been studied in continuous time [23, 8, 5] and discrete time [16, 32], and finite-particle convergence rates have been quantified under several assumption sets [3, 33]. Complementary gradient-flow interpretations on probability spaces [19, 8], in analogy with Jordan, Kinderlehrer, and Otto [14], make explicit that the geometry induced by the kernel affects the size and direction of each descent step.

Kernel selection is therefore a primary quantitative design choice in SVGD: it controls the pairwise particle interaction, the KSD value being reduced, and the extent to which KSD convergence is informative about weak convergence (see also Figures 1 and 9) [11]. The median heuristic [12] is a useful and widely reported baseline, but its scalar bandwidth has no general theoretical optimality guarantee and has been observed to lose accuracy as dimension grows [2, 40, 36]. Several recent methods improve high-dimensional behavior by changing other parts of the algorithmic pipeline [7, 37, 10, 21], and neural variants learn the update direction itself [17, 39]. Our comparison isolates a different source of gains: the numerical effect of replacing the fixed heuristic kernel rule by an adaptive KSD-maximizing kernel choice while leaving the base SVGD transport step intact.

The closest comparison point is Ai et al. [1], which defines a multiple-kernel KSD for a weighted linear combination of base kernels and learns the weights by maximizing that discrepancy. This work supports the same quantitative principle—larger KSD values identify more effective descent directions—but the comparison class is a finite kernel basis. We instead optimize continuous kernel parameters, allowing the empirical comparisons below to test whether adaptive bandwidths alone can produce visible gains over the median-heuristic baseline.

1.2 Contributions.

This paper asks a focused quantitative question: how much of SVGD’s empirical performance gap can be closed by changing only the kernel-selection rule? We deliberately keep the underlying particle transport mechanism unchanged and compare the median heuristic against a principled adaptive alternative, so that differences in Wasserstein error, variance recovery, MMD, and prediction accuracy can be attributed to the kernel choice rather than to architectural or algorithmic enhancements.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We propose a kernel-selection rule that dynamically updates kernel parameters by maximizing the KSD during SVGD inference. Compared with the median heuristic, which supplies one scalar bandwidth, the method optimizes multiple continuous bandwidths and therefore gives the particle update a sharper, dimension-aware discrepancy signal.
- (ii) **Theoretical Analysis.** We analyze the method in the discrete-time mean-field setting by extending fixed-kernel SVGD convergence results. The resulting guarantees are framed around the quantitative object optimized by the algorithm: the supremum of the KSD over the parameterized kernel class converges to zero, and under a Stein logarithmic Sobolev inequality we obtain iteration-complexity bounds in the mean-field limit.
- (iii) **Empirical Validation.** Across the numerical comparisons, we report metrics that directly expose the practical differences between methods—Wasserstein error, marginal variance, co-variance trace, prediction accuracy, and MMD²—and show that adaptive kernel selection consistently improves over the median heuristic while reducing variance collapse.

2 Mathematical background

We fix notation for the quantitative discrepancies used throughout the analysis and experiments. Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a symmetric positive definite kernel with associated RKHS $\mathcal{H}_0$. We write $\mathcal{H}$ for the d -fold Cartesian product of $\mathcal{H}_0$, equipped with $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$, and use the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. The notation $x \cdot y$ denotes the Euclidean inner product and $\nabla \cdot$ the divergence operator. We denote probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ by $\mathcal{P}(\mathbb{R}^d)$ and measures with finite p -th moment by $\mathcal{P}_p(\mathbb{R}^d)$. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance used in the empirical comparisons is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] measure separation between two distributions by taking the largest difference in test-function expectations over a prescribed function class. Stein's method [35] makes this comparison usable when the target distribution is known only up to normalization, because it supplies test functions with zero target expectation. In particular, for $\mathcal{S}_{\pi} f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, $\int_{\mathbb{R}^d} \mathcal{S}_{\pi} f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. This yields the numerical discrepancy optimized in our method, the *kernelized Stein discrepancy* (KSD):

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_{\pi} f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The maximizing direction is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$, where $\mathcal{A}_{\pi}^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with k . Thus the KSD has the concrete value $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$, making it both a convergence metric and the scalar objective used to compare candidate kernels.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial reference distribution μ_0 , SVGD constructs a sequence of approximations by setting $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ is the push-forward under $T(x) = x + \gamma \psi^{\mu_n}(x)$. For a fixed kernel, $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$ is the RKHS steepest-descent direction. The key quantitative identity links this update direction to the KSD: the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes KL divergence. For this choice, the instantaneous KL decrease is exactly governed by the squared KSD:

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these maps to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the finite-particle update used as the common base step in all method comparisons:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to prove without restrictive assumptions, recent analyses often quantify progress through the KSD [16, 33, 32]. This metric is informative but kernel-dependent: KSD decrease by itself need not imply weak convergence if the mean-field measures fail to be tight, and the RKHS in Equation (1) determines how strongly KSD control translates into distributional control [11]. Thus, the same particle system can receive different discrepancy values under different kernels, making kernel comparison a natural part of the convergence and empirical-performance story.

We therefore formulate Ad-SVGd over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, KSD_θ denotes the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the associated optimal update direction, and Φ_θ the feature map. This parameterization turns kernel choice into an explicit optimization variable, so each iteration can compare candidate kernels by the same scalar quantity that controls the instantaneous descent rate.

The selection rule is to choose, or approximate, the kernel that maximizes the KSD between the current empirical particle distribution and the target. The quantitative motivation is immediate from Equations (2) and (7): at a fixed particle measure, the squared KSD determines the size of the first-order KL decrease available to SVGD. Selecting the largest KSD in the class therefore selects the largest certified descent signal among the compared kernels. Geometric analyses of SVGD as a gradient flow support the same ordering, showing that kernels inducing larger KSD values can yield more favorable convergence behavior when their RKHSs are compared [28, 8].

Where fixed-kernel SVGD analyzes the decay of one KSD, Ad-SVGd targets the largest KSD over the kernel class. In contrast to the median heuristic of Liu and Wang [18], the rule applies to any parameterized family: at iteration n , the ideal comparison selects

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

In practice we approximate this comparison by taking one or more gradient-ascent steps with step size $s > 0$ on the KSD before the particle update, and we optionally skip parameter updates on selected SVGD steps to control runtime (see Algorithm 1). The binary decision variable `paramupdate` records whether the kernel comparison is performed at that iteration. The transport step itself can be replaced by variants such as adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates without changing the kernel-selection mechanism; likewise, gradient ascent on the kernel parameter can be replaced by another optimizer. The implementation uses the explicit KSD representation [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

which makes the kernel-parameter gradient available directly from the same pairwise quantities used in the SVGD step. The additional cost relative to vanilla SVGD is therefore concentrated in the kernel-parameter optimization, with two practical mitigations:

- The kernel-parameter ascent reuses the gradients $\nabla \log \pi(X_n^i)$ computed for the corresponding SVGD step, so Ad-SVGD does not add extra evaluations of $\log \pi$.
- Runtime can be reduced by using a small number of kernel steps `nstepstheta`, updating the kernel only after several particle steps, or approximating the empirical KSD with a particle subsample when M is large.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$
Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to nstepstheta  $- 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider targets with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$ and use the convergence analysis to formalize the same comparison principle used empirically. Starting from the fixed-kernel KSD analysis of [32] under Talagrand’s inequality, which is weaker than the commonly used logarithmic Sobolev inequality, we make the assumptions uniform over the kernel parameters. This lets the descent bound compare the selected adaptive kernel against the whole parameter class at each iteration.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

We write the KL objective as

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and recall the fixed-kernel inequality that will serve as the baseline comparison. For a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, with $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ satisfying $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 of [32] gives the following one-step KL upper bound:

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. Applying this inequality to the fixed-kernel SVGD iterates (3) yields the quantitative descent condition [32, Theorem 3.2]

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The critical verification is the step-size condition $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, obtained from Talagrand's inequality in Assumption 2, which permits (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then implies

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Thus (7) is the fixed-kernel baseline against which the adaptive maximal-KSD decrease is compared.

With an adaptive kernel sequence (θ_n) , the same bound tracks the KSD of the kernel actually selected at iteration n :

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

First consider the idealized adaptive update in which the kernel comparison is solved exactly:

$$\begin{aligned}\mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),\end{aligned}\tag{9}$$

so the chosen update direction attains the largest KSD in the class:

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

This exact formulation is available when the KSD maximization over the kernel parameter has a closed-form solution, as in the multiple-kernel SVGd framework [1], where the kernel is a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For continuous or otherwise more general parameter families, the maximizer in (9) is typically approximated; throughout this exact statement, we assume $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The fixed-kernel descent bound from [32] then immediately converts into a maximal-KSD descent statement:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The practical algorithm only approximately solves the KSD maximization, so the comparison must allow an optimization gap. Suppose $\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

where $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ denotes the inner optimizer used to increase $\operatorname{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is gradient ascent on the KSD, although any optimizer with the same asymptotic gap control could be substituted. The required quantitative condition is that the selected kernel eventually matches the best KSD in the class up to a vanishing error.

Assumption 4. We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this approximate comparison condition, the adaptive method still controls the maximal KSD.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32], equivalently applying the fixed-kernel descent result of [32] to the selected parameter, we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and is therefore Cauchy. Thus the one-step decrease, the selected-kernel counterpart of (7), forces

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, the same adaptive comparison can be translated from maximal-KSD convergence to KL convergence for the mean-field system (10). For several regimes of the optimization gap ε_n , Section B gives iteration-complexity bounds and explicitly quantifies the residual bias when KSD maximization is solved only to fixed $\bar{\varepsilon}$ accuracy at each iteration. Importantly, the condition need only hold for one kernel in the class $\{k_{\theta}\}$, which preserves flexibility while still providing a quantitative route from the adaptive KSD objective to KL control.

5 Numerical Experiments

We evaluate Ad-SVGD through direct comparisons with SVGD using the median heuristic (Med-SVGD), reporting metrics that separate mean accuracy, uncertainty recovery, and distributional discrepancy. The main experiments include a one-dimensional Gaussian mixture and a linear inverse problem governed by an ordinary differential equation. Additional dimensional-scaling studies appear in Section A, and Section 5.4 compares the two methods on the Bayesian logistic regression benchmark used in prior SVGD work.

5.1 Kernel parameterization

The baseline kernel family in SVGD is typically

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. The scalar h is the kernel *bandwidth*, and the median heuristic baseline sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med the current median pairwise particle distance. This rule is designed to balance the two terms in the SVGD update (3) [18], but it supplies only one global scale for all coordinates.

For Ad-SVGd, we exploit the parameterized comparison by using product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with coordinate-wise bandwidths $h = (h_1, \dots, h_d)$. This gives the adaptive method a separate scale parameter for each marginal direction while keeping the derivatives required by Algorithm 1 explicit through Equations (4) and (5). We also tested a naive coordinate-wise median heuristic for this product family, but it did not provide a competitive comparison point because it produced variance collapse.

The same quantitative selection rule can be paired with more scalable kernel families. Spectral or low-rank parameterizations could compare dominant variability directions without the full cost of product kernels, while Matérn or mixed-product constructions would allow the KSD objective to tune smoothness and anisotropy in addition to bandwidth.

5.2 Toy example

We first revisit the one-dimensional example from Liu and Wang [18], a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. For each particle count $M \in \{50, 200, 500\}$, particles are initialized from $\mathcal{N}(0, 1)$ and fixed-bandwidth SVGD is run for 10^4 steps with step size 1, using the $p = 1$ kernel family described above. Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit one-dimensional formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) against an exact sample of size 10^5 . Figure 1 shows a pronounced bandwidth window: fixed-bandwidth SVGD reaches low error only for a restricted range of h , while poorly scaled bandwidths give substantially worse Wasserstein distances. This sensitivity provides the first empirical motivation for replacing a static bandwidth rule with an adaptive comparison.

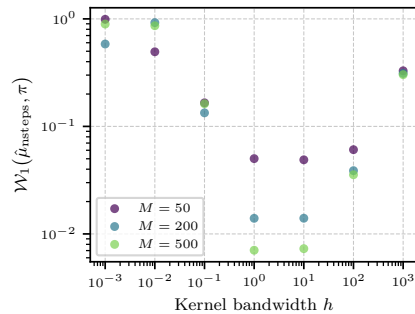


Figure 1: Final Wasserstein 1-distance comparison for the one-dimensional mixture using fixed-bandwidth SVGD across bandwidths h .

5.3 Linear inverse problem based on ODE

The next comparison uses the inverse problem from [38, Example 2.1]. The forward model is the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the inference task is to recover the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation vector is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise. The forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluating the solution at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, the unknown input u receives a Gaussian process (GP) prior represented by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$.

The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For implementation, H is replaced by a numerical discretization of Equation (11) on a grid with mesh size 2^{-8} , and we use the fully observed system with $N_{\text{obs}} = 2^8$. The KL expansion uses $N_x = 16$ terms, the observation-noise covariance is $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$, and reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

The method comparison uses Med-SVGD and Ad-SVGD with particle ensemble sizes $M \in \{50, 100, 200\}$ and the $p = 1$ kernels from Section 5.1. Both methods run for $4 \cdot 10^5$ particle iterations with particle step size 10^{-3} and the AdaGrad-style adaptive step-size control suggested by Liu and Wang [18]. For Ad-SVGD, bandwidth ascent uses step size 10^{-5} and is performed once every 100 particle updates; under this schedule, the observed runtime is not significantly different from Med-SVGD. Figure 2 shows a representative GP reconstruction. The mean estimates from both methods are close, but the 90% uncertainty band highlights the key quantitative difference: Ad-SVGD tracks posterior uncertainty, whereas Med-SVGD contracts the variance. To aggregate this comparison,

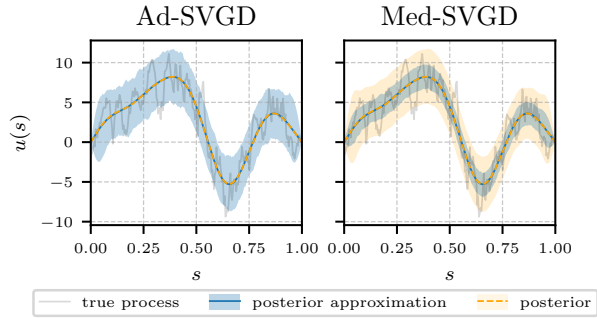
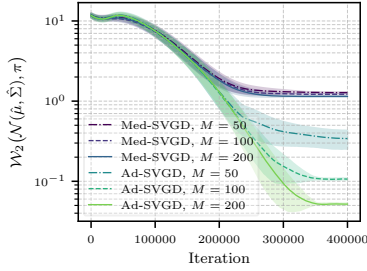


Figure 2: ODE inverse-problem GP reconstruction comparing posterior mean and 90% confidence interval from the particle approximation.

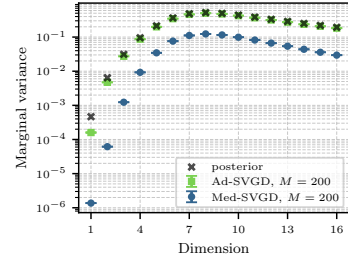
we report the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the true posterior π and the Gaussian approximation defined by the particle sample mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [29, Section 3.2]. Since π is multivariate normal, the metric is computed by the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the posterior mean and covariance. We also compare the final marginal variances against the posterior marginals. Aggregated over 56 random seeds in Figure 3, Ad-SVGD attains the more favorable posterior approximation on both the Wasserstein-error curve and the marginal-variance comparison, while Med-SVGD systematically underestimates uncertainty. The particle-count trend is also favorable to the adaptive method: unlike Med-SVGD, Ad-SVGD improves when the ensemble size increases beyond 50. Figure 4 explains this pattern at the parameter level: the learned component-wise bandwidths stabilize before the approximation error, differ substantially across coordinates, and are negatively aligned with the corresponding marginal variances.

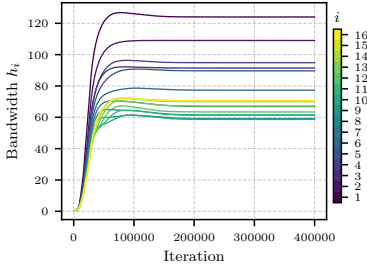


(a) Wasserstein approximation error over iterations

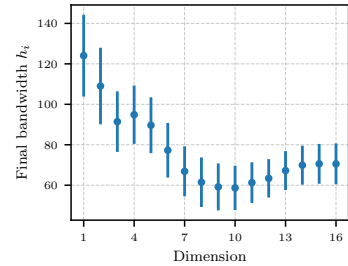


(b) Final marginal-variance comparison

Figure 3: ODE inverse-problem comparison of Med-SVGD and Ad-SVGD, reporting mean and 95% confidence interval over 56 random seeds for approximation error and marginal variance.



(a) Mean bandwidth trajectory by coordinate



(b) Final coordinate-wise bandwidths with mean and 95% confidence interval

Figure 4: Ad-SVGD bandwidth behavior in the ODE inverse problem, aggregated over 56 random seeds to show coordinate-wise stabilization and final variability.

5.4 Bayesian logistic regression

Following Liu and Wang [18] and Liu et al. [21], we compare methods on Bayesian logistic regression (BLR) for the Covertypes data set [4]. The regression weights satisfy $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$

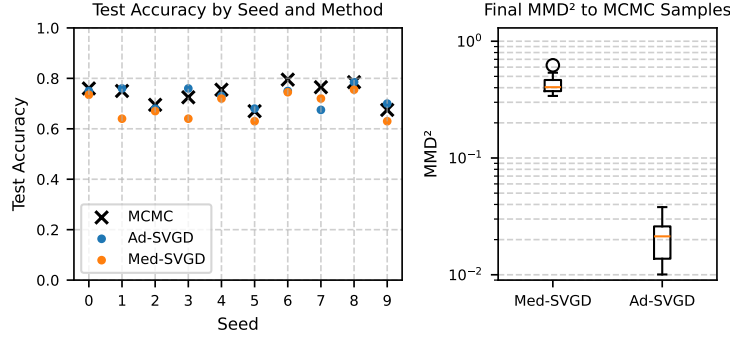


Figure 5: Bayesian logistic regression comparison: prediction accuracy from the particle mean (left) and MMD^2 to NUTS reference samples (right).

with $\alpha \sim \text{Gamma}(1, 0.01)$, and the posterior variable is $x = [w, \alpha]$. To ground the distributional comparison, we follow Liu et al. [21] and use the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] as the reference sampler for evaluating Med-SVGD and Ad-SVGD. As in Liu et al. [21], each run uses a subset of 1000 observations, and results are aggregated over 10 random draws. Classification performance is measured by test-set prediction accuracy using the particle mean. In Figure 5, Ad-SVGD is above the Med-SVGD baseline for almost all seeds in prediction accuracy and is simultaneously closer to the MCMC reference under the reported discrepancy metric.

For posterior approximation quality, the right panel also reports squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. The gap between methods is visible in both summary metrics and second-order structure. Figure 7 compares the final-particle covariance matrices with the MCMC reference, averaged over the 10 seeds using the visualization style of Liu et al. [21]; Med-SVGD underestimates posterior uncertainty, whereas Ad-SVGD is closer to the reference covariance pattern. Figure 6 shows that this improvement is accompanied by substantial bandwidth adaptation: starting from the uninformed value 1 in every component, Ad-SVGD learns coordinate-wise bandwidths that differ from initialization by several orders of magnitude.

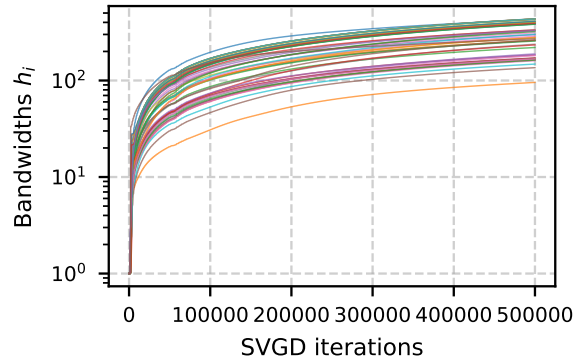


Figure 6: Coordinate-wise Ad-SVGD bandwidth evolution for the BLR experiment.

6 Conclusion

The experiments and analysis support a consistent quantitative framing of SVGD kernel choice: the kernel should be compared by the discrepancy signal it provides at the current particle distribution, rather than fixed by a scalar heuristic. Ad-SVGD implements this principle by ascending the KSD

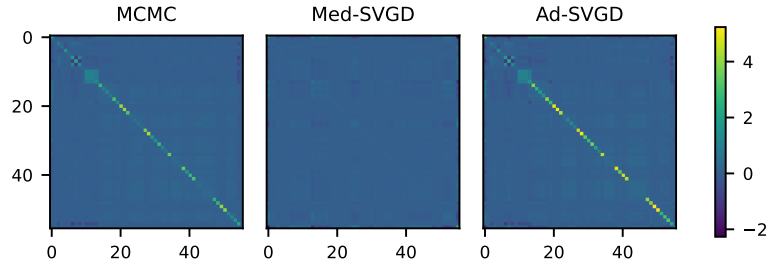


Figure 7: Final-particle covariance matrices for Med-SVGD and Ad-SVGD compared with MCMC reference covariance, averaged over 10 BLR seeds.

over a parameterized kernel family and then applying the standard SVGD particle update. The theory shows that, under the stated assumptions, this adaptive comparison controls the maximal KSD and can be translated into KL bounds under a generalized Stein log-Sobolev condition. Empirically, the same mechanism improves the reported metrics where kernel choice matters most: it matches the median heuristic in the simple one-dimensional mixture, lowers approximation error in higher-dimensional Gaussian tests, better preserves marginal variances and covariance traces, and improves the BLR comparison in prediction accuracy and MMD². These results indicate that adaptive kernel selection is a practical way to recover substantial performance from the original SVGD dynamics without changing the underlying particle transport rule.

7 Limitations and outlook

The main theoretical limitation is the reliance on Assumption 4, which quantifies how closely the inner kernel update approaches the best KSD value in the class. Ongoing work studies when the alternating gradient-ascent scheme in Algorithm 1 satisfies Assumption 4. The present experiments do not prove Assumption 4 for every run, but they show that the implemented update produces consistent practical gains in the reported metrics.

Our analysis intentionally focuses on the original SVGD dynamics so that the empirical margins can be attributed to kernel selection. A natural next step is to combine the same KSD-based adaptive comparison with stronger SVGD variants, including sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27], and then measure whether the gains from adaptive kernels are additive to those algorithmic improvements.

References

- [1] Qingzhong Ai et al. “Stein Variational Gradient Descent with Multiple Kernels”. In: *Cognitive Computation* 15.2 (2023), pp. 672–682. DOI: [10.1007/s12559-022-10069-5](https://doi.org/10.1007/s12559-022-10069-5).
- [2] Jimmy Ba et al. “Understanding the Variance Collapse of SVGD in High Dimensions”. In: *International Conference on Learning Representations*. 2022.
- [3] Krishna Balasubramanian, Sayan Banerjee, and Promit Ghosal. “Improved Finite-Particle Convergence Rates for Stein Variational Gradient Descent”. In: *The Thirteenth International Conference on Learning Representations*. 2025.
- [4] Jock Blackard. *Coverttype*. UCI Machine Learning Repository. 1998. DOI: [10.24432/C50K5N](https://doi.org/10.24432/C50K5N).
- [5] Sinho Chewi et al. “SVGD as a kernelized Wasserstein gradient flow of the chi-squared divergence”. In: *Advances in Neural Information Processing Systems* 33. 2020.
- [6] Kacper Chwialkowski, Heiko Strathmann, and Arthur Gretton. “A Kernel Test of Goodness of Fit”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [7] Gianluca Detommaso et al. “A Stein variational Newton method”. In: *Advances in Neural Information Processing Systems* 31. 2018.
- [8] Andrew Duncan, Nikolas Nüsken, and Lukasz Szpruch. “On the geometry of Stein variational gradient descent”. In: *Journal of Machine Learning Research* 24 (2023).
- [9] Rémi Flamary et al. “POT: Python Optimal Transport”. In: *Journal of Machine Learning Research* 22 (2021).
- [10] Wenbo Gong, Yingzhen Li, and José Miguel Hernández-Lobato. “Sliced Kernelized Stein Discrepancy”. In: *International Conference on Learning Representations*. 2021.
- [11] Jackson Gorham and Lester Mackey. “Measuring Sample Quality with Kernels”. In: *Proceedings of the 34th International Conference on Machine Learning*. 2017.
- [12] Arthur Gretton et al. “A Kernel Two-Sample Test”. In: *Journal of Machine Learning Research* 13 (2012).
- [13] Matthew D. Hoffman and Andrew Gelman. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15 (2014).
- [14] Richard Jordan, David Kinderlehrer, and Felix Otto. “The Variational Formulation of the Fokker–Planck Equation”. In: *SIAM Journal on Mathematical Analysis* 29.1 (1998), pp. 1–17. DOI: [10.1137/S0036141096303359](https://doi.org/10.1137/S0036141096303359).
- [15] Rahif Kassab and Osvaldo Simeone. “Federated Generalized Bayesian Learning via Distributed Stein Variational Gradient Descent”. In: *IEEE Transactions on Signal Processing* 70 (2022), pp. 2180–2192. DOI: [10.1109/TSP.2022.3168490](https://doi.org/10.1109/TSP.2022.3168490).
- [16] Anna Korba et al. “A Non-Asymptotic Analysis for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems* 33. Vol. 33. 2020.
- [17] Lauro Langosco di Langosco, Vincent Fortuin, and Heiko Strathmann. “Neural Variational Gradient Descent”. In: *Fourth Symposium on Advances in Approximate Bayesian Inference*. 2022.

- [18] Q. Liu and D. Wang. “Stein Variational Gradient Descent: A General Purpose Bayesian Inference Algorithm”. In: *Advances in Neural Information Processing Systems 29*. 2016.
- [19] Qiang Liu. “Stein Variational Gradient Descent as Gradient Flow”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [20] Qiang Liu, Jason Lee, and Michael Jordan. “A Kernelized Stein Discrepancy for Goodness-of-fit Tests”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [21] Xing Liu et al. “Grassmann Stein Variational Gradient Descent”. In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*. 2022.
- [22] Yang Liu et al. “Stein Variational Policy Gradient”. In: *Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence*. 2017.
- [23] Jianfeng Lu, Yulong Lu, and James Nolen. “Scaling Limit of the Stein Variational Gradient Descent: The Mean Field Regime”. In: *SIAM Journal on Mathematical Analysis* 51.2 (2019), pp. 648–671. DOI: [10.1137/18M1187611](https://doi.org/10.1137/18M1187611).
- [24] Safa Messaoud et al. “S2AC: Energy-Based Reinforcement Learning with Stein Soft Actor Critic”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [25] Krikamol Muandet et al. “Kernel Mean Embedding of Distributions: A Review and Beyond”. In: *Foundations and Trends in Machine Learning* 10.1-2 (2017), pp. 1–141. DOI: [10.1561/22000000060](https://doi.org/10.1561/22000000060).
- [26] Alfred Müller. “Integral Probability Metrics and Their Generating Classes of Functions”. In: *Advances in Applied Probability* 29.2 (1997), pp. 429–443. DOI: [10.2307/1428011](https://doi.org/10.2307/1428011).
- [27] Nikolas Nüsken. *Stein transport for Bayesian inference*. 2024. arXiv: [2409.01464](https://arxiv.org/abs/2409.01464).
- [28] Nikolas Nüsken and D. R. Michiel Renger. “Stein variational gradient descent: Many-particle and long-time asymptotics”. In: *Foundations of Data Science* 5.3 (2023), pp. 286–320. DOI: [10.3934/fods.2022023](https://doi.org/10.3934/fods.2022023).
- [29] Victor M. Panaretos and Yoav Zemel. “Statistical Aspects of Wasserstein Distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: [10.1146/annurev-statistics-030718-104938](https://doi.org/10.1146/annurev-statistics-030718-104938).
- [30] Yuchen Pu et al. “VAE Learning via Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [31] Sebastian Reich and Simon Weissmann. “Fokker–Planck Particle Systems for Bayesian Inference: Computational Approaches”. In: *SIAM/ASA Journal on Uncertainty Quantification* 9.2 (2021), pp. 446–482. DOI: [10.1137/19M1303162](https://doi.org/10.1137/19M1303162).
- [32] Adil Salim, Lukang Sun, and Peter Richtarik. “A Convergence Theory for SVGD in the Population Limit under Talagrand’s Inequality T1”. In: *Proceedings of the 39th International Conference on Machine Learning*. 2022.
- [33] Jiaxin Shi and Lester Mackey. “A Finite-Particle Convergence Rate for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 36*. 2023.
- [34] Björn Sprungk, Simon Weissmann, and Jakob Zech. “Metropolis-adjusted interacting particle sampling”. In: *Statistics and Computing* 35.64 (2025). DOI: [10.1007/s11222-025-10595-w](https://doi.org/10.1007/s11222-025-10595-w).

- [35] Charles Stein. “A bound for the error in the normal approximation to the distribution of a sum of dependent random variables”. In: *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 2: Probability Theory*. 1972, pp. 583–603.
- [36] Dilin Wang, Zhe Zeng, and Qiang Liu. “Stein Variational Message Passing for Continuous Graphical Models”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.
- [37] Dilin Wang et al. “Stein Variational Gradient Descent With Matrix-Valued Kernels”. In: *Advances in Neural Information Processing Systems 32*. 2019.
- [38] Simon Weissmann, Ashia Wilson, and Jakob Zech. “Multilevel Optimization for Inverse Problems”. In: *Proceedings of Thirty Fifth Conference on Learning Theory*. 2022.
- [39] Qian Zhao et al. “Stein variational gradient descent with learned direction”. In: *Information Sciences* 637 (2023). DOI: [10.1016/j.ins.2023.118975](https://doi.org/10.1016/j.ins.2023.118975).
- [40] Jingwei Zhuo et al. “Message Passing Stein Variational Gradient Descent”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.

A More numerical experiments

A.1 Gaussian mixture models

We revisit the mixture experiment from Section 5.2 to separate two comparisons: fixed bandwidth sensitivity and adaptive versus median bandwidth selection. For the fixed-bandwidth setup, Figure 9 compares the final particle histograms with the target density for $h \in \{0.001, 1, 1000\}$. We then compare Med-SVGD and Ad-SVGD across particle counts M , always using the final particle distribution after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance from the final empirical particle distribution to the target for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. In this low-dimensional setting, both curves improve with particle count and converge to essentially the same accuracy regime, with Wasserstein distance below 0.01 at $N = 500$. Thus, the adaptive method preserves the strong one-dimensional performance of the median heuristic rather than trading it away.

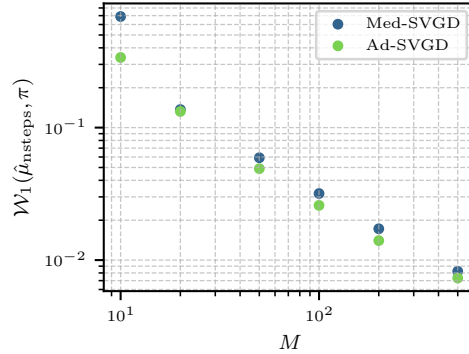


Figure 8: Wasserstein 1-distance comparison for the one-dimensional Gaussian mixture as particle count varies.

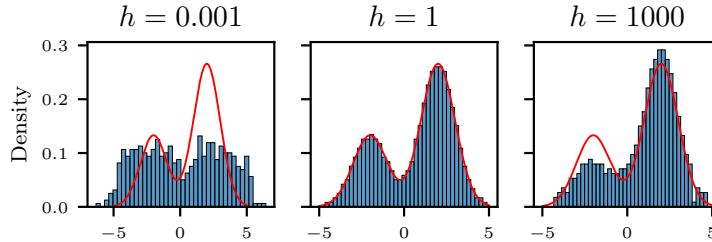


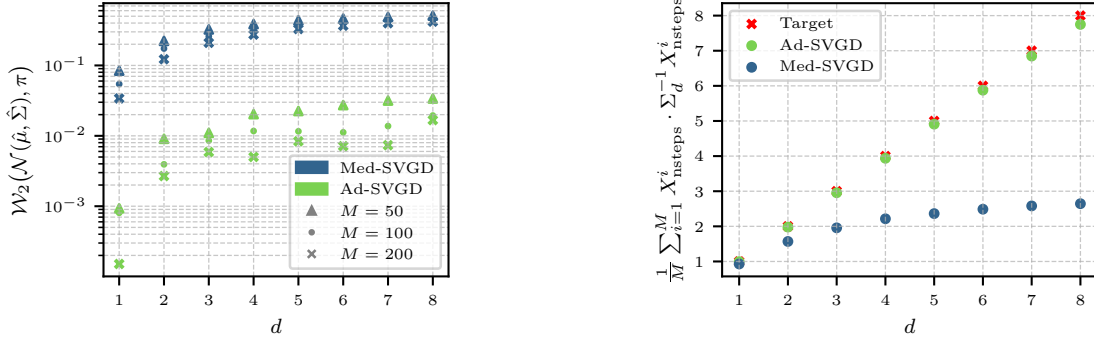
Figure 9: Final-particle histograms for fixed-bandwidth SVGD with $h = 0.001, 1, 1000$; the red curve is the target density π , highlighting the bandwidth-dependent approximation quality.

Overall, the one-dimensional mixture is a near-tie between the adaptive and median rules: both approximate the target well once enough particles are available. This result is useful as a baseline because it shows that Ad-SVGD does not lose accuracy in the regime where the median heuristic is already well matched to the problem.

A.2 Scaling dimension: multivariate normal distribution

To test whether the comparison changes with dimension, we next use Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and both algorithms are run for 10^4 iterations with step size 0.1. When numerical stability required a smaller step size, the number of iterations was adjusted accordingly so that the comparison remained method-focused rather than stability-limited.

Approximation quality is evaluated from the final particles as dimension increases. Since the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see [29]), we compare π_d with $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particle set [29, Section 3.2]. Following the terminology of the Python Optimal Transport routine used for the calculations [9], we call this the *Bures Wasserstein distance*. Figure 10a reports this metric for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$; across every tested dimension d , the adaptive bandwidth comparison gives the lower approximation error.



(a) Bures Wasserstein approximation error for $M \in \{50, 100, 200\}$. (b) χ^2 uncertainty statistic for $M = 200$ compared with its target expectation.

Figure 10: Multivariate-normal scaling comparison: Ad-SVGD and Med-SVGD are evaluated by Bures Wasserstein error and a χ^2 uncertainty statistic as dimension increases.

Because the Bures Wasserstein distance can downweight errors in the low-variance coordinates of π_d , we also report the uncertainty-sensitive statistic $X \cdot \Sigma_d^{-1} X$. For $X \sim \mathcal{N}(0, \Sigma_d)$, this statistic is χ^2 -distributed with d degrees of freedom and therefore has expectation d (cf. [34, p. 27]). We compute its particle average for $M = 200$ and compare it with d in Figure 10b. As dimension grows, Med-SVGD moves increasingly far from the expected value, whereas Ad-SVGD remains comparatively close, making the variance-retention advantage visible in a scalar diagnostic. The marginal-variance tables give the component-level version of the same comparison for $M = 200$. Tables 1 and 2 list the target marginal variances in the first row and the particle variances for each d below. Med-SVGD consistently undershoots the target variances, with the deficit becoming especially clear in higher dimensions in Table 1. Ad-SVGD, by contrast, stays close to the target variance in every component in Table 2, matching the favorable χ^2 pattern in Figure 10b. Finally, we visualize the full marginal distributions for the most challenging reported case, $M = 200$ and $d = 8$. Each coordinate is rescaled by the inverse target marginal standard deviation, equivalently multiplying the i -th component by i , so the target marginal becomes standard normal. Figures 11 and 12 show the resulting histograms and Figures 13 and 14 show the corresponding Q-Q plots. The visual comparison is consistent with the numerical tables: Ad-SVGD follows the standard-normal marginals closely across components, while Med-SVGD displays the same underdispersion seen in the variance metrics.

Table 1: Med-SVGD marginal variances for final particles with $M = 200$ across dimensions d ; the target variances of π_d are shown in the first row for direct comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Ad-SVGD marginal variances for final particles with $M = 200$ across dimensions d ; the target variances of π_d are shown in the first row for direct comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

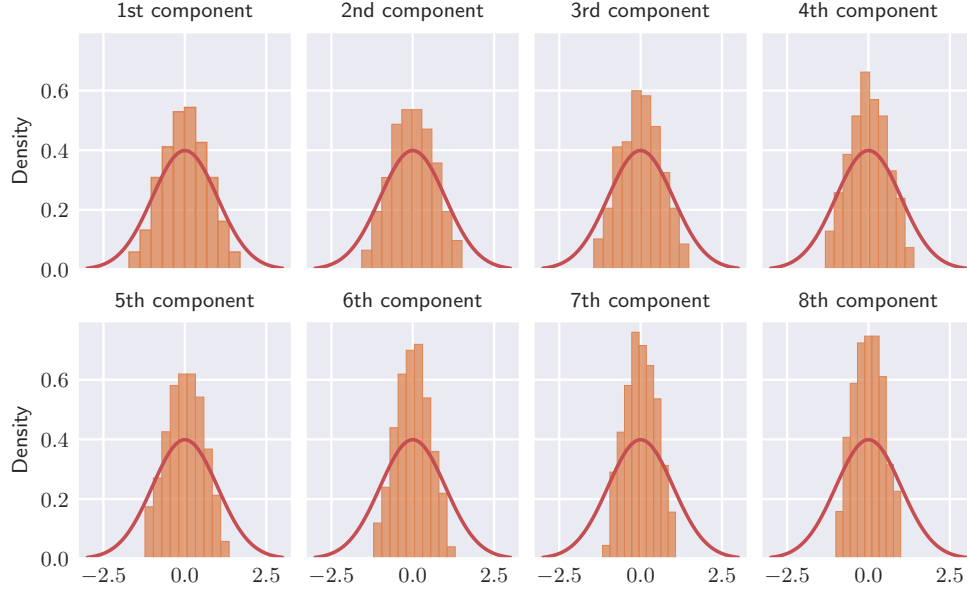


Figure 11: Med-SVGd marginal histograms for the eight-dimensional case with $M = 200$, after rescaling by target marginal variances; the standard Gaussian density provides the target reference.

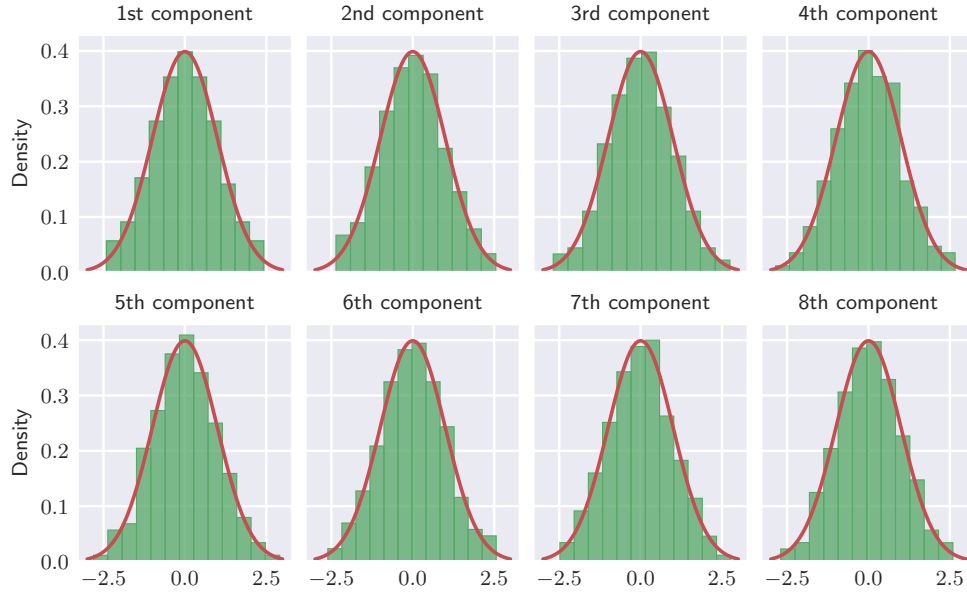


Figure 12: Ad-SVGd marginal histograms for the eight-dimensional case with $M = 200$, after rescaling by target marginal variances; the standard Gaussian density provides the target reference.

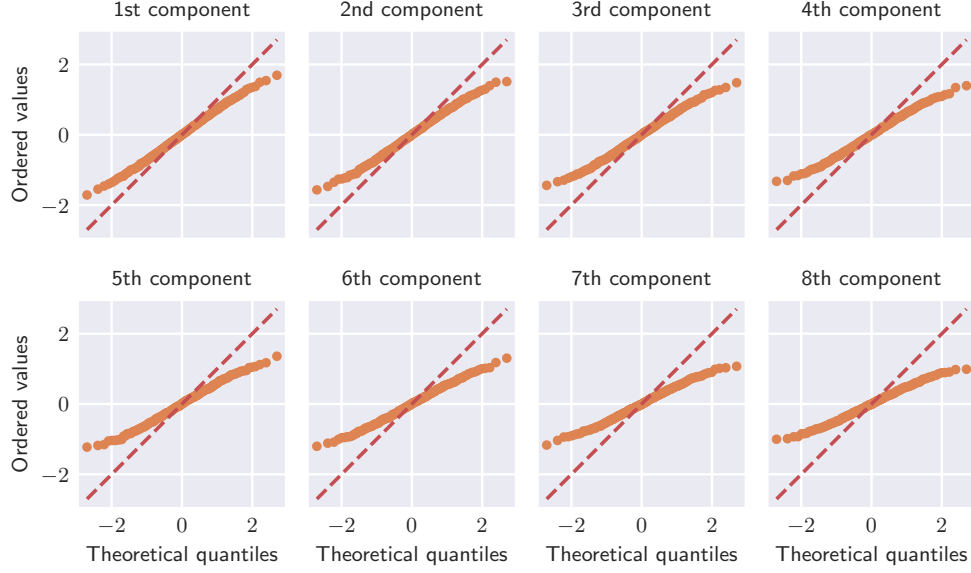


Figure 13: Med-SVGd Q-Q comparison for rescaled marginals in the eight-dimensional case with $M = 200$; the unit-slope line through the origin marks agreement with the standard normal target.

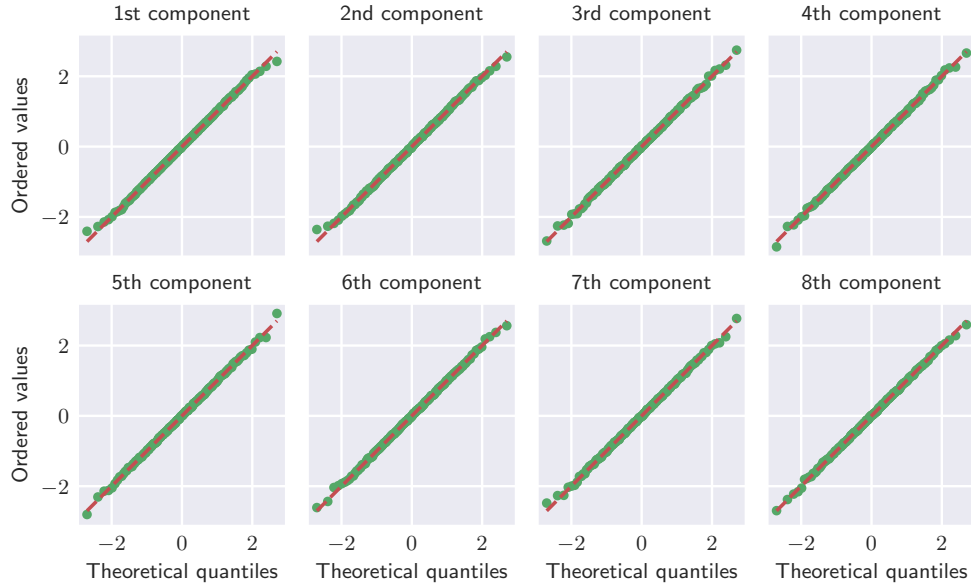


Figure 14: Ad-SVGd Q-Q comparison for rescaled marginals in the eight-dimensional case with $M = 200$; the unit-slope line through the origin marks agreement with the standard normal target.

A.3 Gaussian process inference

We further test dimensional scaling on the Gaussian process (GP) inference task proposed in [31]. The GP on $[0, 1]$ is represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ has independent Gaussian coordinates $x_k \sim \mathcal{N}(0, k^{-2})$. Observations are taken at N_y equispaced points in $[0, 1]$, and the goal is to infer the coefficients x_k . For fixed N_x and N_y , the inverse problem has forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ and $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal entries k^{-2} , $k = 1, \dots, N_x$, the noise is $\varepsilon \sim \mathcal{N}(0, I_{N_y})$, and reference observations are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

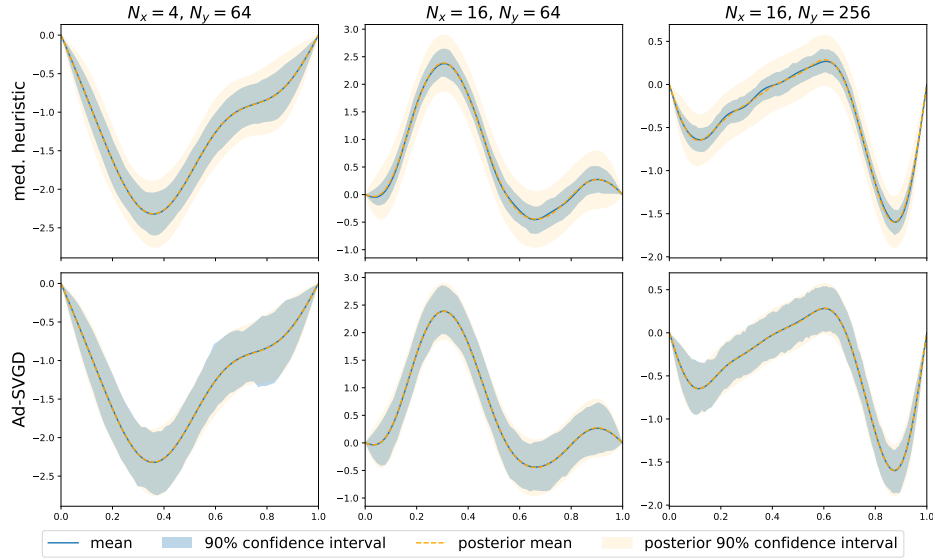


Figure 15: Posterior-process comparison for median-heuristic SVGD and Ad-SVGd with $M = 100$ particles across model configurations.

We sample from the posterior $X | Y = \bar{y}$ and compare the median heuristic with Ad-SVGd under matched particle count. Med-SVGd uses $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas Ad-SVGd uses the product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with coordinate-wise bandwidths $h = (h_1, \dots, h_{N_x})$. Following [18], an Adagrad variant controls step sizes when $N_x = 16$.

Figure 15 compares the resulting process estimates with the posterior for $N_x = 4, N_y = 64$, $N_x = 16, N_y = 64$, and $N_x = 16, N_y = 256$. The visual pattern is consistent across observation counts: as coefficient dimension increases, Med-SVGd contracts the posterior band, while Ad-SVGd remains closer to the posterior uncertainty. Table 3 quantifies the same effect with covariance traces. In every listed configuration, the Ad-SVGd trace is much closer to the theoretical posterior trace than the median-heuristic trace, indicating substantially better variance recovery.

B Refined theoretical analysis

The main convergence statement can be sharpened when a Stein log-Sobolev inequality links KL divergence to KSD. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Covariance-trace comparison for final particle distributions with $M = 100$ particles: theoretical posterior, median heuristic, and Ad-SVGD, averaged over 25 runs.

conditions under which it may hold. In the adaptive setting, the quantitative requirement is less restrictive than asking every kernel to satisfy the inequality: because Ad-SVGD compares kernels by the maximum KSD over the class, it is enough for one kernel in the class to provide the log-Sobolev control. We formalize this comparison through the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property converts the maximal-KSD descent from the main analysis into a KL recursion. In particular, it yields an explicit bound on $\mathcal{F}(\mu_n)$ even when the inner KSD maximization has only a uniform fixed accuracy $\varepsilon_n \leq \bar{\varepsilon}$, rather than a vanishing error.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}$$

for all $n \in \mathbb{N}$.

Proof. As in the proof of Theorem 3, Assumption 5 and the conditions on α, γ give a one-step comparison between KL decrease, the maximal KSD, and the kernel-optimization error:

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

The three stated regimes follow by bounding this weighted error sum separately:

1. If $\varepsilon_n \leq \bar{\varepsilon}$, the fixed residual term is controlled by the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1-\rho}.$$

2. If $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$, the KL error obeys

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1-\rho/q} = q^n \frac{1}{q-\rho}$$

using the geometric-series formula. In the complementary case $q < \rho$, the same calculation gives

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1-q/\rho} = \rho^n \frac{1}{(\rho-q)}.$$

When $q = \rho$, replacing q^k by $\tilde{q}^k$ for any $q < \tilde{q} < 1$ yields the stated rate by the same comparison.

3. For polynomial optimization error $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we split the weighted sum into recent and earlier terms:

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor+1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here the first half of the sum uses $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$, while the remaining terms use $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□